כריית טקסט - קובץ פלט לתרגיל מספר 2

מגישים:

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סעיף 3

דוגמא לביקורות לא מדויקות והסבר :

א. ביקורת חיובית שקיבלה ציון שלילי – קיימות מספר מילים שליליות כמו :
"couldn't" – מילה זו בדרך כלל מתארת שלילה, אך כאן היא חלק מביטוי חיובי
"just" – תלוי בהקשר, מילה זו יכולה להיות מרמזת על משהו מוגבל או מצמצם, אבל כאן היא מתארת קרבה חיובית.

Comment: The location is awesome!! Its just a couple of blocks from South Station, close to great restaurants, and bars and the .VIEW!! couldn't have picked a better location. The Boston Tea Party Museum is close too

Calculated Sentiment: 0.3 -0.02357023 -0.3265986 0

Average Sentiment: -0.01309019

ב. ביקורת חיובית שקיבלה ציון שלילי :
Quiet-למרות שההקשר כאן חיובי (שקט), ייתכן שבסיטואציות אחרות השקט ייראה כחיסרון, למשל אם מחפשים מקום תוסס.
could have spent much more days - ייתכן שמישהו יפרש את זה כחצי ביקורת על כך שהייתם מוגבלים בזמן ולא הצלחתם לנצל את השהות

Comment: We spent 5 nights in Sanjay's appartement. For the two of us it had the perfect size. We could have spent much more days there! All in all a really modern, quiet and clean apartment.

Calculated Sentiment: -0.1020621 0.2371708 -0.8980256 0.6577538

Average Sentiment: -0.0262907

סעיף 4

ChatGPT prompt:

Please write this text like this: The location is awesome!! Its just a couple of blocks from South Station, close to great restaurants, and bars and the VIEW!! couldn't have picked a better location. The Boston Tea Party Museum is close too. Classify the sentence as positive/negative or neutral. And write the sentiment score.

תוצאה:

Rewritten Text:
The location is incredible! Just a short walk from South Station, near amazing restaurants, bars, and THE VIEW!! Absolutely perfect. Plus, the Boston Tea Party Museum is nearby too.

Classification:

Positive

Sentiment Score:

Sentiment scores typically range from -1 (very negative) to +1 (very positive). This sentence would likely score +**0.95** due to its highly enthusiastic and positive tone.

גירסת הצ׳אט: הגרסה הנוכחית של הצ׳אט היא GPT-4.

:ChatGPT prompt

Please write this text like this: We spent 5 nights in Sanjay's appartement. For the two of us it had the perfect size. We could have spent much more days there! All in all a really modern, quiet and clean apartment. Classify the sentence as positive/negative or neutral. And write the sentiment score.

תוצאה :

Rewritten Text:
We stayed for 5 nights at Sanjay's apartment. It was the perfect size for the two of us. We could have easily spent even more time there! Overall, a very modern, quiet, and clean place.

Classification:

Positive

Sentiment Score:

+0.90 — The tone is highly positive, reflecting satisfaction and enthusiasm about the experience.

גירסת הצ'אט : הגרסה הנוכחית של הצ'אט היא GPT-4.

סעיף 5

מספר שורות מהקובץ כולל שני חישובי הסנטימנט (צילום מסך של המשפטים המכילים את המילה/הביטוי שהוספתם ללקסיקון):

```
num: 18
Comment:  jose was always available. feedback and directions were great - fantastic location. local corner coffee shop for brea
kfast... excellent. salem street right around the corner. lots of stairs to clime, th floor - don't bring lots of luggage. bed
was quite comfortable. apartment needs to be cleaner. bathroom has seen its better days, needs repairs.
Average Sentiment:  0.3849909
Average Sentiment Updated:  0.2690138


num: 19
Comment:  listing accurate, clean and well kept apartment. neighborhood lively and conveniently located. owner was extremely he
lpful and available for questions.
Average Sentiment:  0.8181407
Average Sentiment Updated:  0.9441289


num: 20
Comment:  thoroughly enjoyed our time in this wonderful house and had the pleasure of meeting robin & victoria. had never done
airbnb before and the concept of letting yourself in and making yourself at home seemed a little strange to begin with, however
robin kept us informed via text. the area is great for eating and exploring and has a real community spirit. transport links ar
e also good. would definitely stay here again.
Average Sentiment:  0.3553985
Average Sentiment Updated:  0.4390213


num: 21
Comment:  great apartment location and very spacious. ideal for exploring boston. only downside is its and older apartment and
could be cleaner.
Average Sentiment:  0.497384
Average Sentiment Updated:  0.3466284


num: 22
Comment:  the room was not shown as advertised. the room and bathroom seemed much smaller. there was no kitchen sink, one of th
e towels provided still smelled like bo, the shower had mold growing on the tiles and the bathroom was much smaller. the bed an
d comforter provided had stains on them. it resembled a college dorm room. the only good thing about our stay was that we did n
ot have to stay in that room for more than hours. the area it self was nice. close to the train and northeastern university.
Average Sentiment:  0.04821567
Average Sentiment Updated:  0.02140263
```

סעיף 6

Comparison cloud :

מונחים :

5 מונחים חיוביים בולטים: great, boston, comfortable, clean, recommend

5מונחים שליליים בולטים : Room, Dirty, however, night, like

```
# Using libraries

library(stringi)

library(textstem)

library(tm)

library(qdap)

library(sentimentr)

library(wordcloud)

library(ggthemes)

library(lexicon)

# set working directory

setwd("C:/Users/niv/Desktop/Text mining/EX2")

#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.1~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

# Set Options

options(stringsAsFactors=FALSE) #don't treat strings as factors (categories)

Sys.setlocale('LC_ALL','C')


# read csv

airbnb.df<-read.csv('bos_airbnb_1k.csv')
```

```
head(airbnb.df)

#text coloum into new data frame
comments.df<-airbnb.df[,'comments']
head(comments.df)

#Create corpus
comments.vec<- VectorSource(comments.df)
comments.cor<- VCorpus(comments.vec)
class(comments.cor)

# Cleaning test by function
check <- function(text.cor,name) {
 cat(name,"\n\n")
 for (i in 1:6) {
  print(text.cor[[i]][1])
 }
 cat("\n\n")
```

```
}

### Text cleaning

# transferring data to lowercase
comments.cor  <- tm_map(comments.cor,content_transformer(tolower))
check(comments.cor,"lowercase")

# remove numbers
comments.cor  <- tm_map(comments.cor , removeNumbers)
check(comments.cor,"remove numbers")

# remove white spaces
comments.cor <- tm_map(comments.cor ,stripWhitespace)
check(comments.cor,"remove white spaces")

comments.cor <- lemmatize_words(comments.cor)
```

```r
# Convert cleaned corpus to text
cleaned_comments <- unlist(lapply(comments.cor, function(x) x$content))


#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.2~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
# Sentiment analysis
airbnb.df$sentiment <- sentiment_by(cleaned_comments,question.weight=0)$ave_sentiment
head(airbnb.df$sentiment)
table(sign(airbnb.df$sentiment))


#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.3~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
for (i in 1:nrow(airbnb.df)) {
 # Check if the sentiment is within the specified range
 if (airbnb.df$sentiment[i] < 0 && airbnb.df$sentiment[i] > -0.1) {
  # Calculate the sentiment for the current comment
  senti <- sentiment(cleaned_comments[i],question.weight=0)$sentiment

  # Initialize positive and negative sentiment accumulators
  positive <- 0
```

```
negative <- 0

for (var in senti) {
  if (var > 0) {
    positive <- positive + var
  } else {
    negative <- negative + var
  }
}

# Check if the sum of positive and negative sentiments meets criteria
if ((positive + negative >= -0.15) && (positive != 0)) {
  # Print the 'comments' column for the current row
  cat("Comment: ", airbnb.df$comments[i], "\n")

  # Print the calculated sentiment value
  cat("Calculated Sentiment: ", senti, "\n")
```

```
      # Print the average sentiment value from the data frame
      cat(″Average Sentiment： ″, airbnb.df$sentiment[i], ″\n″)


      # Add a blank line for better readability
      cat(″\n″)
    }
  }
}



#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.4~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#ChatGPT：
#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.5~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~


#check if a word exists in the lexicon
lexicon：：hash_sentiment_jockers_rinker[″dirty ″] #NA


lexicon：：hash_sentiment_jockers_rinker[″repair″] #NA
```

```
lexicon::hash_sentiment_jockers_rinker["be cleaner"] #NA

lexicon::hash_sentiment_jockers_rinker["kept"] #NA

lexicon::hash_sentiment_jockers_rinker["mold"] #NA

# add new words to the dictionary
library(lexicon)
updated_hash_sentiment <- sentimentr:::update_polarity_table(lexicon::hash_sentiment_jockers_rinker,
                                        x = data.frame(
                                          words = c('be cleaner', 'repairs','dirty','mold','kept'),
                                          polarity = c(-1, -1,-1,-1,1),
                                          stringsAsFactors = FALSE
                                        )
)
```

```
airbnb.df$sentiment_updated <-
sentiment_by(cleaned_comments,polarity_dt=updated_hash_sentiment,question.weight=0)$ave_sentiment

head(airbnb.df$sentiment_updated)

table(sign(airbnb.df$sentiment_updated))


# writhe to csv

write.csv(airbnb.df,'bos_airbnb_1k_sentiment_new.csv')


new_words <- c('be cleaner', 'repairs', 'dirty', 'mold', 'kept')

num<-1

# Loop through each row in the DataFrame

for (i in 1:nrow(airbnb.df)) {

  # Extract comments for the current row

  comments <- cleaned_comments[i]


  # Loop through each comment if it's a list or vector of comments

  for (j in 1:length(comments)) {

    # Loop through the new_words

    bool<-TRUE
```

```
    for (k in 1: length(new_words)) {
     # Check if the new_word appears in the comment
     if (grepl(new_words[k], comments[j], ignore.case = TRUE)&&bool&&airbnb.df$sentiment[i]!= airbnb.df$sentiment_updated[i]) {
      bool<-FALSE
      cat("num: ",num,"\n")
      num<-num+1
      cat("Comment: ", comments[j], "\n")
      cat("Average Sentiment: ", airbnb.df$sentiment[i], "\n")
      cat("Average Sentiment Updated: ", airbnb.df$sentiment_updated[i], "\n")
      cat("\n\n")
     }
    }
   }
  }
```

```
#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.6~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

#Dividing positive and negative comments
negative.df<-airbnb.df[airbnb.df$sentiment<0,'comments']
positive.df<-airbnb.df[airbnb.df$sentiment>0,'comments']

#cleaning
clean.vec<-function(text.vec){
  text.vec <- tolower(text.vec)
  text.vec<-removeWords(text.vec,stopwords('english'))
  text.vec <- removePunctuation(text.vec)
  text.vec <- removeNumbers(text.vec)
  text.vec <- stripWhitespace(text.vec)
}

positive.vec<-clean.vec(positive.df)
negative.vec<-clean.vec(negative.df)
```

```
#concatenate all the comments together
negative.vec<-paste(negative.vec,collapse = " ")
positive.vec<-paste(positive.vec,collapse = " ")

comparisonCloud.df<-c(positive.vec,negative.vec)
comparisonCloud.corpus<-VCorpus(VectorSource(comparisonCloud.df))
tdm<-TermDocumentMatrix(comparisonCloud.corpus)
tdm.matrix<- as.matrix(tdm)
colnames(tdm.matrix)= c("positive","negative")

comparison.cloud(tdm.matrix,max.words = 150,random.order = FALSE,title.size = 1,
          colors = brewer.pal(ncol(tdm.matrix),"Dark2"))
```